

Experimental Demonstration of Compilation-oriented AI Agent for Autonomous PON Slice Provisioning

Chansung Park⁽¹⁾, Yongwook Ra⁽¹⁾, HwanSeok Chung⁽¹⁾

⁽¹⁾ Optical Network Lab., Electronics and Telecommunications Research Institute (ETRI), 218 Gajeong-ro, Yuseong-Gu, Daejeon, 34129, South Korea, [chansung18, nyw0242, chung}@etri.re.kr](mailto:{chansung18, nyw0242, chung}@etri.re.kr)

Abstract We demonstrate a compilation-oriented AI agent for autonomous PON slicing. By aggregating multi-source evidence into a shared specification, the AI agent enables model-agnostic improvements in provisioning reliability and token efficiency. Experiments on a 25G PON testbed demonstrate consistent dual-slice provisioning performance across multiple LLMs. ©2026 The Author(s)

Introduction

Passive optical networks (PONs) are increasingly expected to support services with heterogeneous bandwidth and latency requirements, ranging from conventional broadband traffic to time-sensitive mobile transport. In this context, network slicing enables multiple service groups to share a common physical access infrastructure while maintaining isolation in resource allocation, traffic treatment, and performance objectives. Although prior studies have shown the feasibility of sliced optical access [1–5], a key practical challenge remains in translating high-level operational intent into executable provisioning procedures based on operational information from multiple system views.

To address this challenge, this paper presents a compilation-oriented AI agent for autonomous PON slice provisioning. The agent compiles high-level service intent and operational evidence from multiple system views into an explicit and executable service specification, which is progressively instantiated at runtime and maintained as a shared state throughout the provisioning process. Experiments on a real 25G PON testbed demonstrate that our agent improves provisioning success rates and reduces token usage across multiple LLMs, while correctly provisioning both mobile and business slices.

System Architecture

The proposed AI agent integrates a compilation engine into a closed-loop control architecture to handle the complexity of PON slice provisioning, as shown in Fig. 1. A basic agent loop interprets an operator request, execute control actions, and determine subsequent steps based on immediate feedback [6, 7]. While effective for localized tasks, the conventional loop lacks a persistent context required for PON slice provisioning, as the required operational evidence is progressively resolved.

The compilation engine addresses this gap by consolidating runtime evidence into a service engineering specification, which serves as the shared state of the orchestration. The agent consults and

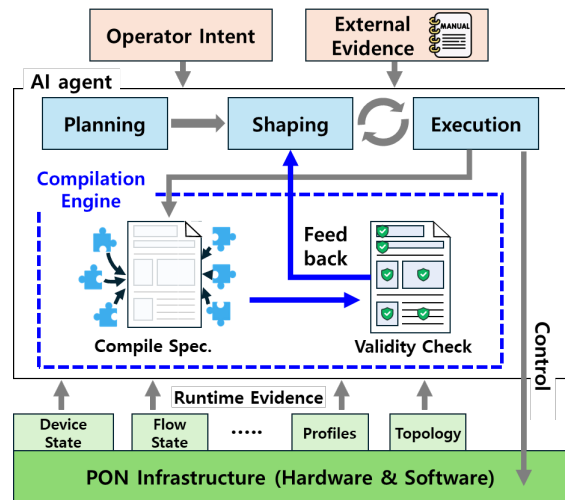


Fig. 1: Compilation-oriented AI agent integrating compilation engine into a conventional closed-loop control architecture.

updates this specification across planning, execution, and validation. Its formal schema is given in the following section.

Service Engineering Specification

Each provisioning request is represented by an instance of a service engineering specification that serves as the explicit shared state of the orchestration. This specification provides a unified executable view by consolidating runtime facts that are otherwise scattered across multiple operational sources. The schema is fixed across requests, while individual field values are progressively instantiated from operational evidence collected at runtime. Formally, we denote such an instance as

$$S = (I, C, T, M, B, A, V), \quad (1)$$

where I represents request identity, C control context, T transport binding, M member and service evidence, B bandwidth and policy constraints, A actuation state, and V verification results. In practical terms, these groups capture the target request, the control-plane context required to reach the target path, the PON-side transport bindings, the admitted subscriber and service evidence, the compiled slice budget and service mode, the actuation

state, and the corresponding read-back evidence. Together, these groups define the runtime state shared between orchestration and control.

During execution, the specification is updated exclusively through evidence returned by executed actions. Let e_k denote the evidence obtained at step k . The update process is written as

$$\begin{aligned} S_{k+1} &= \Phi(S_k, e_k), \quad k < K, \\ K &= \min\{k \mid E(S_k) = 1\}, \end{aligned} \quad (2)$$

where Φ updates only the fields whose prerequisites are satisfied by the newly obtained evidence, while leaving unresolved fields unchanged, and where $E(S_k)$ denotes the executability condition of the specification at step k . In this way, the framework progressively completes the specification instance using runtime evidence until the earliest step at which the request becomes executable, thereby enabling the agent to operate on a structured and auditable state rather than relying solely on an ever-growing dialog history.

The specification also carries the slice-level bandwidth condition. Let U denote the admitted member set for the target slice. The configured slice bandwidth must be sufficient to accommodate the admitted demand while remaining within the residual capacity of the PON channel. The admissibility condition is

$$\sum_{u \in U} (B_{\text{fixed},u} + B_{\text{assured},u}) \leq B_{\text{cfg}} \leq C_{\text{budget}}, \quad (3)$$

where $B_{\text{fixed},u}$ and $B_{\text{assured},u}$ represent the fixed and assured bandwidths allocated to member u , respectively. B_{cfg} denotes the configured slice bandwidth, and C_{budget} the residual bandwidth budget within the usable bandwidth of the PON channel capacity (i.e., excluding overhead). Runtime validation then checks whether the system is ready, whether the resolved path is consistent,

whether the admitted members belong to the intended service group, and whether the compiled slice budget satisfies the condition above. Once these checks succeed, the specification serves both as the control input for actuation and as an auditable record of the provisioning result.

Testbed and Integrated Workflow

The experimental testbed consists of a G-PON-based white-box 25G OLT and two 25G ONUs, operating at 25 Gb/s downstream and 10 Gb/s upstream. To emulate mobile services, the setup includes DU and RU emulators, each equipped with 10 GbE interfaces. An IEEE 1588 PTP grandmaster clock (GMC) is inserted between the DU emulator and the OLT to enable synchronized co-operative DBA (Co-DBA) operation. An optical splitter connects the OLT and ONUs, forming the optical distribution network.

The software stack is based on the ONF SEBA architecture [8] and customized for PON slice provisioning. VOLTHA (Virtual OLT Hardware Abstraction) provides an abstraction layer and management interface for the underlying physical PON, while ONOS hosts control-plane applications, including AAA for subscriber authentication, SADIS for subscriber and bandwidth profile management, and vOLT for service flow control. For performance monitoring (PM) of mobile service latency, the testbed employs an Elasticsearch–Grafana stack. PM data are collected by a JSON exporter that retrieves OLT register values via VOLTHA and exports them in JSON format for indexing and visualization. Closed-source LLMs, including GPT-5.4, Claude Opus 4.6, and Claude Sonnet 4.6, are accessed via service APIs, whereas open-weight models (gemma-4-31b-it, Qwen3.5-27B, and Qwen-3.5-35B-A3B) are deployed using vLLM on an ASUS ESC8000A-E12 server equipped with $8 \times \text{H100}$ GPUs. Fig. 2 illustrates the complete ex-

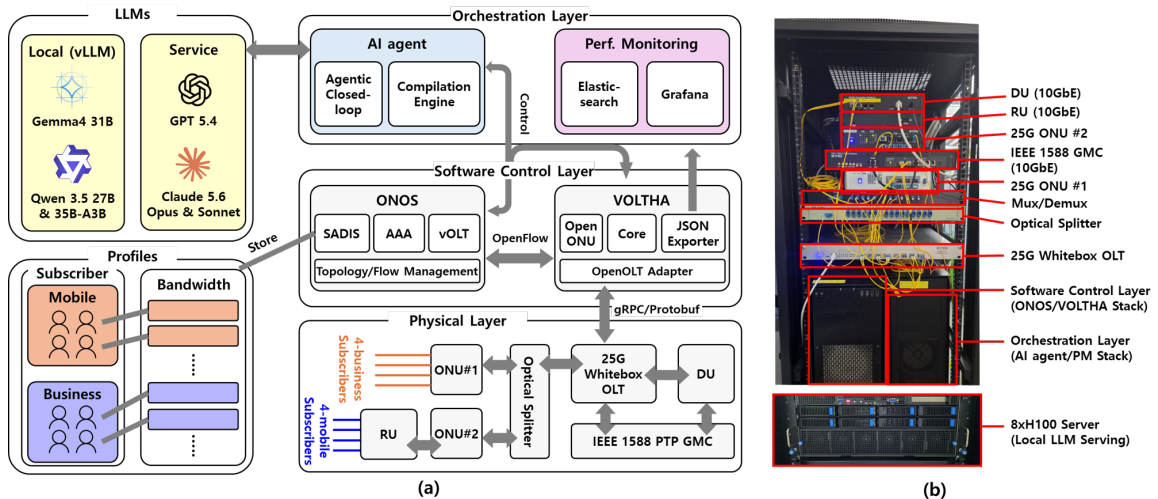


Fig. 2: Experimental environment for PON slice provisioning: (a) logical architecture of the control and orchestration, including SEBA-based components and AI agent integration; (b) physical hardware configuration of the 25G PON testbed.

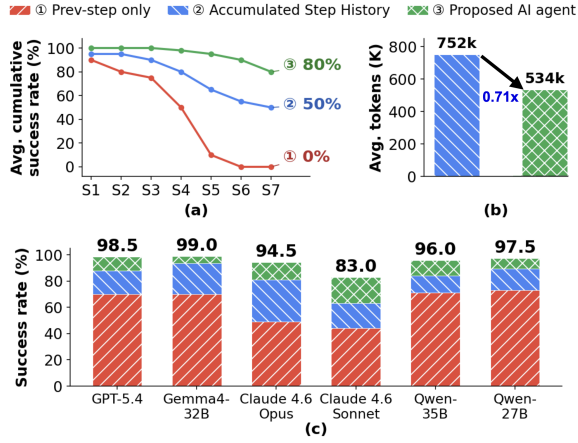


Fig. 3: Performance comparison of orchestration strategies: (a) stage-wise provisioning success rate, (b) averaged token usage, and (c) model-wise success rates across LLMs.

perimental environment, including both the logical architecture and the physical deployment.

The provisioning workflow is evaluated as a single integrated procedure rather than a set of isolated subtasks. To facilitate the evaluation, four business and mobile subscribers are predefined on ONU 1 and ONU 2 in SADIS, respectively. Each subscriber profile specifies the subscriber identifier, VLAN tags, a technology profile ID, and upstream and downstream bandwidth profiles. The bandwidth profile consists of B_{fixed} , B_{assured} , and best-effort (BE), defining guaranteed aggregate rates such that the total upstream allocation, including BE traffic, does not exceed the available bandwidth budget.

Performance of compilation-oriented AI agent

We first examine the integrated provisioning workflow using GPT-5.4 as the reference model. In Fig. 3(a), S1–S7 denote the seven provisioning stages, from OLT-state resolution to flow-status verification; the workflow completes all stages without rollback while progressively consolidating runtime evidence into a service specification.

To evaluate whether explicit compilation provides advantages over baseline approaches, we compare a reactive agent relying on immediate feedback, a context-accumulating agent that propagates conversational history, and the compilation-oriented architecture. Fig. 3(a) and (b) show that the architecture exhibits substantially less stage-wise performance degradation, leading to a higher overall completion rate and more reliable execution than the baselines, while also requiring fewer tokens, suggesting a reduction in redundant reasoning overhead. Token efficiency is defined as the total number of tokens consumed during the end-to-end provisioning workflow. Fig. 3(c) further shows that this advantage remains consistent across different LLMs, indicating that the explicit specification of intermediate states provides a structured shared context that reduces ambiguity

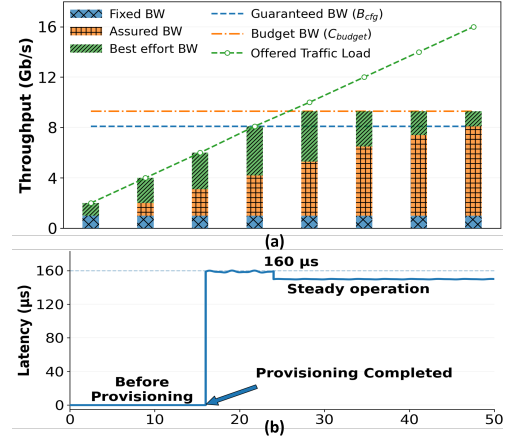


Fig. 4: QoS and latency performance of the provisioned slices: (a) upstream bandwidth allocation under varying traffic load, and (b) latency behavior of the mobile slice after provisioning.

and limits error propagation during orchestration.

QoS and latency of provisioned slices

To verify QoS enforcement for the eight subscribers across ONU 1 and ONU 2, we evaluate upstream bandwidth allocation under varying traffic loads, as shown in Fig. 4(a). Given the 10 Gb/s capacity limit of the 25G OLT port, the system operates within strict bandwidth constraints. The results show that the network prioritizes the guaranteed bandwidths (B_{fixed} and B_{assured}), maintaining the guaranteed bandwidths up to their configured limits as the offered load increases, while adapting best-effort traffic accordingly. This behavior confirms correct QoS policy enforcement under constrained conditions.

In addition, Fig. 4(b) shows that the latency of the mobile slice stabilizes after provisioning, remaining below approximately 160 μs . This behavior demonstrates stable low-latency operation, indicating successful activation of the Co-DBA mechanism and validating both control-plane correctness and service-level performance [9, 10].

Conclusion

We presented a compilation-oriented AI agent for autonomous PON slice provisioning. The agent consolidates distributed runtime evidence into an explicit and executable service specification, maintained as a shared state throughout provisioning. Experiments on a real 25G PON testbed showed improved provisioning success rates, reduced token usage across multiple LLMs, and correct provisioning of both mobile and business slices. These results demonstrate the practical value of explicit specification-based orchestration for reliable and efficient AI-driven optical access systems.

Acknowledgements

This work was supported by IITP grant funded by the Korea government(MSIT)[RS-2023-00215959, Development of Access Agnostic wired and wireless integrated optical access technology].

References

- [1] H. Uzawa et al., "First demonstration of bandwidth-allocation scheme for network-slicing-based tdm-pon toward 5g and iot era", in *Optical Fiber Communication Conference (OFC) 2019*, Optica Publishing Group, 2019, W3J.2. DOI: 10.1364/OFC.2019.W3J.2. [Online]. Available: <https://opg.optica.org/abstract.cfm?URI=OFC-2019-W3J.2>.
- [2] C. Centofanti et al., "Slice management in sdn pon supporting low-latency services", in *European Conference on Optical Communication (ECOC) 2022*, Optica Publishing Group, 2022, Tu5.64. [Online]. Available: <https://opg.optica.org/abstract.cfm?URI=EEOC-2022-Tu5.64>.
- [3] C. Centofanti, A. Marotta, D. Cassioli, F. Graziosi, V. Gudepu, and K. Kondepu, "End-to-end slicing via o-ran and software defined optical access", in *Optical Fiber Communication Conference (OFC) 2023*, Optica Publishing Group, 2023, W3F.4. DOI: 10.1364/OFC.2023.W3F.4. [Online]. Available: <https://opg.optica.org/abstract.cfm?URI=OFC-2023-W3F.4>.
- [4] X. Liu et al., "First field trial of llm-powered ai agent for lifecycle management of autonomous driving optical networks", in *Optical Fiber Communication Conference (OFC) 2025*, Optica Publishing Group, 2025, Th1A.2. DOI: 10.1364/OFC.2025.Th1A.2. [Online]. Available: <https://opg.optica.org/abstract.cfm?URI=OFC-2025-Th1A.2>.
- [5] Y. Ra et al., "Field trial of remotely controlled smart factory based on pon slicing and disaggregated olt", in *2022 European Conference on Optical Communication (ECOC)*, 2022, pp. 1–3.
- [6] C. Park, Y. Ra, and H. S. Chung, "Ai-vpon: A trustworthy sim-to-real ai agent for reliable end-to-end pon operations", in *2025 30th Asia-Pacific Conference on Communications (APCC)*, 2025, pp. 1–6. DOI: 10.23919/APCC64555.2025.11279819.
- [7] C. Park, Y. Ra, and H. S. Chung, "Ai-volt: Multi-stage agentic translation from operator intent to executable pon procedures", in *2025 16th International Conference on Information and Communication Technology Convergence (ICTC)*, 2025, pp. 1266–1270. DOI: 10.1109/ICTC66702.2025.11387884.
- [8] T. Suzuki et al., "Demonstration of ieee pon abstraction for sdn enabled broadband access (seba)", *J. Lightwave Technol.*, vol. 39, no. 20, pp. 6434–6442, Oct. 2021. [Online]. Available: <https://opg.optica.org/jlt/abstract.cfm?URI=jlt-39-20-6434>.
- [9] K. Kim, K. Doo, and H. Chung, "Demonstration of low latency 25g tdm-pon with flexible multizone-based onu activation for time critical services", in *2022 European Conference on Optical Communication (ECOC)*, 2022, pp. 1–3.
- [10] J.-Y. Oh, Y.-C. Ryoo, K.-O. Kim, K.-H. Doo, H.-H. Lee, and H.-S. Chung, "Sdn based real-time latency measurement and its monitoring for 5g mobile convergence passive optical networks", in *2019 Asia Communications and Photonics Conference (ACP)*, 2019, pp. 1–3.